

Wallet Sizing Methodology

Imagine staring into a massive, bustling room through a tiny keyhole. That is precisely how Synbank views a client's true financial universe. Through their internal ledgers, they see every transaction that passes through their doors with absolute clarity. But they are looking at a single room in a much larger mansion. The true **Total Wallet Size (TAW)**—the millions flowing to competing institutions—remains hidden in the shadows.

Even in the dark, we aren't walking blind. Financial reality gives us hard physical boundaries to anchor our search:

- **The Unshakable Floor:** Total wallet size can never be less than what Synbank has already captured — a strict lower bound.
- **The financial proxy ceiling:** External financial statements (OpEx, Cost of Goods Sold, Total Revenue) can't trace every dollar, but they cap the absolute upper boundary of what a client could spend.
- **The Fog of Uncertainty:** The same figure often masquerades under different names across filings, and independent sources frequently clash. Rather than forcing consensus, the model widens its interval whenever signals conflict.

1. Model, per product pillar

For a client and a single product pillar $p \in \{\text{Transactional Banking, Investment Banking, Global Markets}\}$, let X_p be that client's total wallet for pillar p over the observed period. The model is fit separately per pillar meaning a signal belongs to exactly one pillar's posterior, never pooled across pillars. With all the constraints and the nature of financial activity, the following off-the-shelf model was chosen.

$$X_p \sim \text{LogNormal}(\mu_p, \sigma_p^2), \quad Y_p = \log X_p \sim \mathcal{N}(\mu_p, \sigma_p^2), \quad \sigma_p^2 \sim \text{Inv-Gamma}(a_0, b_0), \quad \mu_p \mid \sigma_p^2 \sim \mathcal{N}\left(m_0, \frac{\sigma_p^2}{k_0}\right)$$

(m_0, k_0, a_0, b_0) are set from Synbank's own internal ledger for that client, restricted to pillar p .

Assumption: one stationary rate per pillar within the period, with (m_0, k_0, a_0, b_0) only as reliable as the client's own internal ledger history.

2. External evidence: converting $x_1, \dots, x_n$ into wallet units

Each raw external signal E_i — a disclosed metric such as OpEx, Cost of Goods Sold, Total Revenue, an FX gain, or a SENS announcement — is not a wallet figure and cannot enter the model as-is. Two conversions turn it into a usable x_i :

- (i) Metric $\rightarrow$ volume: apply a multiplier c_i appropriate to that metric (e.g. for a fee-driven line at known rate r , $c_i = 1/r$), isolating the volume-driven component from confounds first.
- (ii) Time-scale $\rightarrow$ period scale: if E_i is reported on a different horizon than the observed period P , rescale by P/P_i , the native reporting period of E_i .

$$x_i = \log(E_i \cdot c_i) + \log\left(\frac{P}{P_i}\right)$$

Pillar attribution: x_i only enters pillar p 's pool if the disclosure is specific to that pillar's activity (e.g. FX/derivatives turnover $\rightarrow$ Global Markets only); a client-level figure that cannot be split by pillar is not used as evidence for any single X_p .

Limitation: c_i is treated as known and error-free — a mis-specified multiplier (e.g. FX gains conflating fee income with revaluation) biases x_i silently.

3. Pooling evidence and updating the prior

Deduplicated, converted signals $x_1, \dots, x_n$ for pillar p are pooled at equal reliability. The raw count is discounted for correlation between signals (shared macro drivers, overlapping disclosures) — this is also where disagreement between sources is captured: conflicting signals inflate SS, which widens the final interval:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad SS = \sum_{i=1}^n (x_i - \bar{x})^2, \quad n_{\text{eff}} = \frac{n}{1 + (n-1)\rho}$$

Normal-Inverse-Gamma posterior update, with n_{eff} applied consistently throughout:

$$k_n = k_0 + n_{\text{eff}}, \quad m_n = \frac{k_0 m_0 + n_{\text{eff}} \bar{x}}{k_n}, \quad a_n = a_0 + \frac{n_{\text{eff}}}{2}, \quad b_n = b_0 + \frac{SS}{2} + \frac{k_0 n_{\text{eff}} (\bar{x} - m_0)^2}{2k_n}$$

Assumption: equal reliability across signals ($w_i = 1$) and one shared correlation ρ for all pairwise dependence — simplifications made because only 1–4 signals exist per client.

4. Final output

Point estimate (median, not mean):

$$\hat{X}_{p, \text{median}} = \exp(m_n)$$

Credible interval — respecting the internal floor and the financial-proxy ceiling — via the Student-t predictive:

$$s^2 = \frac{b_n(k_n + 1)}{a_n k_n}, \quad \text{df} = 2a_n, \quad \hat{X}_{p,q} = \exp(m_n + t_{q, 2a_n} \cdot s), \quad [X_{\text{lo}}, X_{\text{hi}}] = [\hat{X}_{p, \alpha/2}, \hat{X}_{p, 1-\alpha/2}]$$

Limitation: the interval reflects only modelled uncertainty (σ^2 , signal disagreement) — not unseen competitor channels or an undetected conversion error.

5. Total wallet size

Total Wallet Size (TAW) sums the pillar posteriors, $\text{TAW} = X_{TB} + X_{IB} + X_{GM}$ with the uncertainty interval combing each pillar's half width

$$w = \frac{X_{hi} - X_{lo}}{2}, \quad [\text{TAW}_{lo}, \text{TAW}_{hi}] = \text{TAW} \pm \sqrt{w_{TB}^2 + w_{IB}^2 + w_{GM}^2}$$

Assumption: pillar draws summed independently (no cross-pillar correlation). Constraint: TAW bounded below by Σ captured, above by Σ proxy ceiling.